

2.2.1 Measurements and uncertainties

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) systematic errors (including zero errors) and random errors in measurements	
(b) precision and accuracy	As discussed in <i>The Language of Measurement</i> (ASE 2010).
(c) absolute and percentage uncertainties when data are combined by addition, subtraction, multiplication, division and raising to powers	As set out in the ASE publication <i>Signs, Symbols and Systematics (The ASE Companion to 16–19 Science, 2000)</i> . A rigorous statistical treatment is not expected. <i>M1.5</i>
(d) graphical treatment of errors and uncertainties; line of best fit; worst line; absolute and percentage uncertainties; percentage difference.	An elementary knowledge of error bars is expected at A level. HSW5 <i>M1.5</i>